Date : 01/02/26

Introduction to Computer Graphics

Rendering : Generate 2D image of a object.

Define computer Graphics.

* Graphics system → Most importance.
* Explain application areas.
* Education and training.

Image processing applies techniques to modify or interpret the existing image.

↳ Symbol to pixel.

Uses of computer Graphics.

- Controllable — Games, Animations.

Non. controllable — phone-book, setting etc.

end product — complete product.

↓

Using C.G create both product.

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Graphics Hardware

Host computer : Can say output device or display device.

→ Cathode Ray Tube (CRT)

→ Vector / Random scan Display.

→ Raster scan display.

CRT is the most common display device today

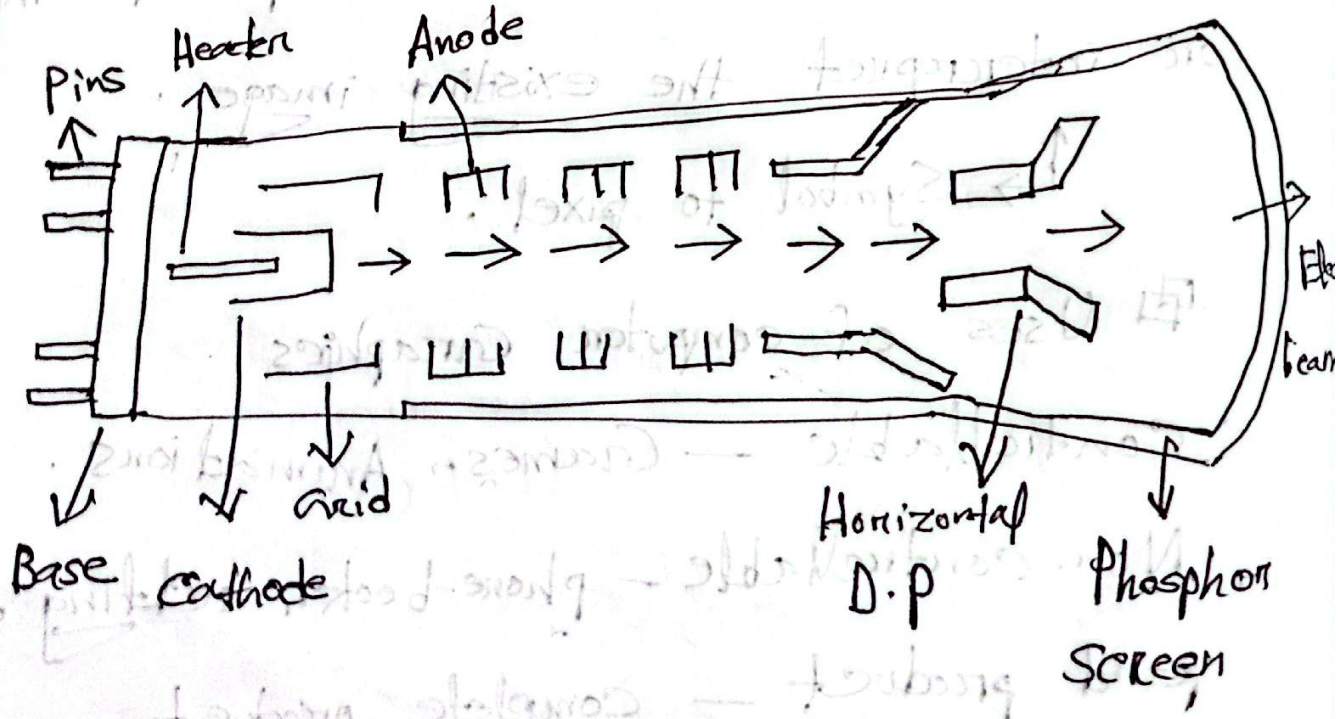

Fig: Cathode Ray Tube

CRT Works :

Works of CRT:

→ It converts electrical signal (voltage, current) to visual signal.

→ The electron gun contains a filament that, when heated, emits a stream of electron.

→ Electrons are focused with an electromagnet into a sharp beam and directed to a specific point of the face of the picture tube.

→ The front surface of the picture tube is coated with small phosphers dots.

→ When the beam hits a phosphere dot it glows with a brightness proportional to the strength of the beam and how long it is hit.

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Color CRT Monitors:

Beam Penetration Method.

Can display only four colors.

Gun & Shape

→ Shadow mask.

Scan Conversion

How to represent a line in a better way

Scanline → Row

~~Definition of scan conversion / masterization~~

~~$P_2(2.2, 1.3)$ Floor → $P_2(2, 1.2)$~~

~~$P_3(2.8, 1.9)$ Floor → $P_3(2, 1.3)$~~

~~$+0.5 \rightarrow (3.3, 2.4)$ Floor → $P_3(3.2)$~~

Definition of scan conversion / Masterization

$P_2(2.2, 1.3)$ Floor → $P_2(2.1)$

$P_3(2.8, 1.9)$ Floor → $P_3(2.1)$

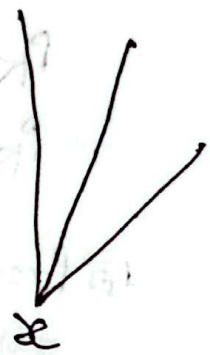
$+0.5 \rightarrow (3.3, 2.4)$ Floor → $P_3(3.2)$

If we draw a line we need. $mx+c=y$

Stairstep appearance

Incremental Fashion

$\tan\theta = 1$



(x_1, y_1)
(x_2, y_2)

$$\frac{(y-y_1)}{(x-x_1)} = \frac{(y_2-y_1)}{(x_2-x_1)}$$

* 1 → Costly operation (disadvantage)

$y = y_1 + 0.5$

$\frac{dy}{dx}$. $y = y+m$

$= y + \frac{dy}{dx}$

if $x+1 = y + \frac{dy}{1}$ that form,

$= y+m$

DDA → Digital Differential Analyzer

Explain of DDA.

Bresenham's Line Algorithm

Why use Bresenhams Line Algorithm

there has a parameter P

Dicision parameter → P

$P_K < 0 = (X_{K+1}, Y_K)$

$P_K \geq 0 = (X_{K+1}, Y_{K+1})$

Prove*

$y = mx + c/b$

$y = X_{K+1}$

$y = m(X_{K+1}) + c/b$

Decision parameter

(x_1, y_1)

(i) $\Delta x = x_2 - x_1$

$\Delta y = Y_2 - Y_1$

(ii) D.P $P_K = 2\Delta Y - \Delta X$

Case I: $X_{K+1} = X_{K+1}$

$Y_{K+1} = Y_K$

$$P_{k+1} = P_k + 2\Delta Y$$

Case II : $P_k \geq 0$

$$X_{k+1} = X_{k+1}$$

$$Y_{k+1} = Y_{k+1}$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x$$

maintain table → লিখতে হবে

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2D

Zoom in/out called scale

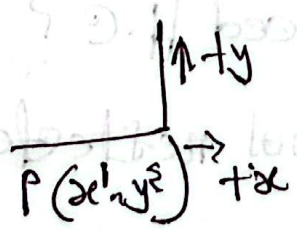
moving - translation

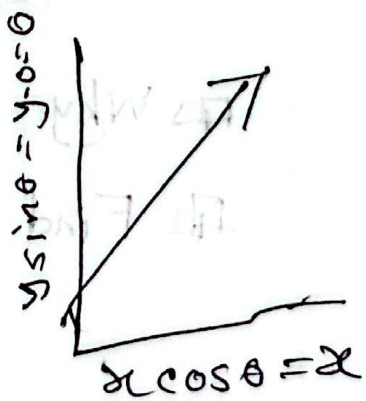
another one y rotate.

$x = x' + tx$

$x' = x - tx$

$y' = y + ty$





Rotation : Positive angle → θ

→ -θ

$x = r\cos\theta$

$y = r\sin\theta$

$x' = x\cos\theta - y\sin\theta$

$y' = x\sin\theta - y\cos\theta$

Translation → 2 matrix add.

Rotation → 2 matrix multiply.

Scale → 2 " "

Homogenous coordination:

$x = \frac{x_h}{h}$ $y = \frac{y_h}{h}$

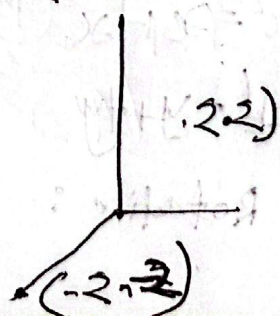
$(x, y) \rightarrow h$

$x = \frac{x_h}{h}$ $y_h = \frac{y_h}{h}$

$x_h = x \cdot h$ $y_h = y \times h$

Why need H.C ?

Find out reflected point on mirror



fix X
↓
reject

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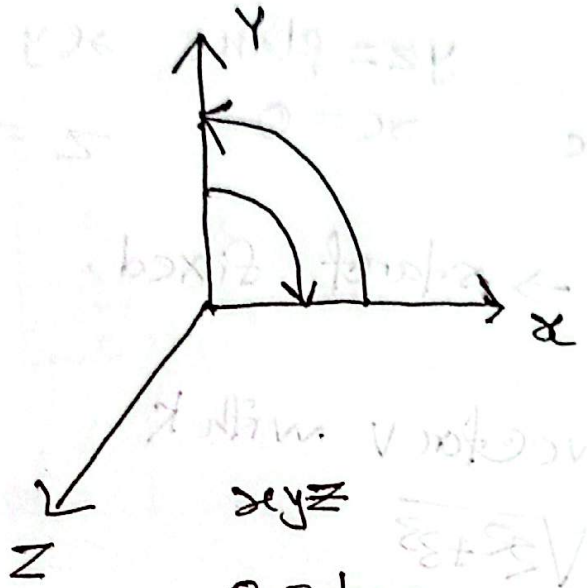
Composite transformation. 3D

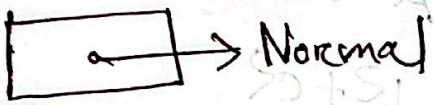
Order of Rotation



xyz

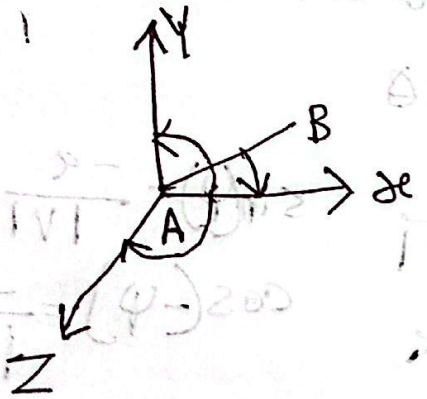
θ = tn-

Order change rotation place change



Inverse Rotation. opposite of Rotation.

x, y, z about Rotation.



rotation about

x/y/z axis

i, j, k

x, y, z

aligning a vector with a vector

Relative scaling

math

$y = 0$ $\quad$ $yz =$ plane $\quad$ xy plane

$xz =$ plane $\quad$ $x = 0$ $\quad$ $z = 0$

anticlock → start fixed.

Aligning vector v with k

$(2,3)$. $\sqrt{2^2+3^2}$

$v = \sqrt{a^2+b^2+c^2}$

$a^2 + l^2 = a^2+b^2+c^2$

$\Rightarrow l^2 = b^2+c^2$

$\therefore l = \sqrt{b^2+c^2}$

$\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

$\cancel{\sin(\varphi) = \frac{-a}{|v|}}$ $\quad$ $\sin(\varphi) = \frac{-a}{|v|}$

$\cancel{\cos(-\varphi) = \frac{l}{|v|}}$ $\quad$ $\cos(-\varphi) = \frac{l}{|v|}$

$$A_V = \begin{bmatrix} \cos(-\phi) & 0 & \sin(-\phi) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\phi) & 0 & \cos(-\phi) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_V = \begin{bmatrix} \frac{\lambda}{|V|} & 0 & \frac{-a}{|V|} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{a}{|V|} & 0 & \frac{\lambda}{|V|} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\lambda}{|V|} & \frac{-ab}{\lambda|V|} & \frac{-ac}{\lambda} & 0 \\ 0 & \frac{c}{\lambda} & \frac{-b}{\lambda} & 0 \\ \frac{a}{|V|} & \frac{b}{|V|} & \frac{c}{|V|} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Alignment:

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$A_V^{-1} = \left(R_{-\phi,i} \times R_{\theta,i}\right)^{-1}$$

$$= R^{-1}_{\theta,i}$$

Aligning vector $v(v)$ with $n(a)$

Transpose:

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}^T = \begin{bmatrix} 0 & 3 & 6 \\ 1 & 4 & 7 \\ 2 & 5 & 8 \end{bmatrix}$$

General 3D rotation:

actual general rotation 5

total 7 [standing translate]

mirror reflection (mcq) matrix

basic 3 সূত্র

Z-axis shear → chapter

rotation matrix

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$z' = (0 \cdot x) + (0 \cdot y) + (1 \cdot z) + (0 \cdot 1)$$

$$= 0 + 0 + z + 0$$

$$= z$$

direct mcq*